GGG-W-665b
October 30, 1963

SUPERSEDING
Int. Fed. Spec. GGG-W-00665a (Navy-Ships)
September 13, 1961 and
Fed. Spec. GGG-W-665
July 12, 1954

FEDERAL SPECIFICATION

WRENCH, SPANNER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers adjustable and nonadjustable spanner wrenches generally used for machinery maintenance and for tightening and loosening hose couplings and hydrant caps.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, grades, and classes.* Spanner wrenches shall be of the following types, grades, and classes as specified (see 6.1):

Type I.—Adjustable hook.
- Class 1.—Fixed pivot point.
- Class 2.—Variable pivot point.
- Class 3.—Heavy duty.

Type II.—Pin.
- Grade A. — Hardened and tempered steel.
- Grade B.—Casehardened steel or malleable iron.

Type III.—Face.
- Class 1.—Adjustable
- Class 2.—Nonadjustable.

Type IV.—Combination hydrant and spanner.

Type V.—Universal, hose coupling.

Type VI.—Double eye.

Type VII.—Nonadjustable hook, for spring grip couplings.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Specifications:

TT-C-490—Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

TT-E-485 — Enamel, Semi-Gloss, Rust-Inhibiting.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 151—Metals; Test Methods.

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-E-15090 — Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are descriptive and not restrictive and included for the convenience of identification by the requisitioning and purchasing offices and manufacturers, and are not intended to preclude the purchase of spanner wrenches which are otherwise in accordance with this specification.

3.2 **Material.** The chemical composition of the materials and the heat-treatment employed shall be such as to produce wrenches complying with the physical requirements specified hereinafter for each type.

3.3 **Design and manufacture.** Wrenches shall be suitable for the purpose intended. Handles shall afford a comfortable grip. Surfaces shall be smooth, and corners shall be broken to approximately 1/32-inch radius. Fillets having a radius of at least 1/32 inch shall be provided.

3.3.1 *Tolerance.* Dimensions referred to herein as approximate shall be interpreted as allowing a plus or minus tolerance of 15 percent from such dimensions.

3.4 **Identification marking.** Wrenches shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.4.1 *Military agencies.* Wrenches shall be marked for identification as specified in MIL-STD-130.

3.5 **Finish.**

3.5.1 *Surface roughness.* All surfaces shall be thoroughly cleaned and free from pits, nodules, forged flash marks, cast flash marks, burrs, cracks, or other detrimental defects. All forge or cast flash marks shall be completely removed to blend smoothly with the adjacent surfaces. Determination with this paragraph shall be made on the finished product (after the required coating is applied).

3.5.2 *Coatings.* Unless otherwise specified (see 6.1), wrenches shall have one of the following coatings (see 6.1) which shall be adherent, smooth, continuous and free from uncoated areas, pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability. The minimum thickness of the coating shall be as specified in 3.5.2.1 through 3.5.2.4 on all surfaces which can be touched by a ball 0.750 inch in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE I. - Type I, class 1, adjustable hook, fixed pivot point, spanner wrenches

Capacity for circles, diameter	Length over all approximate	Thickness		Depth of hook +1/32 -1/64	Test load minimum	Capacity for slotted hose couplings											
		Handle approximate	Hook approximate			Hose diameter						Couplings diameter					
						Hose types 1/						Hose types 1/					
						A	B	C	D	E	F	A	B	C	D	E	F
Inches	Inches	Inch	Inch	Inch	Inch-pounds	Inches	Inches	Inches	Inch	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
3/4 to 2	6 3/8	1/4	11/32	1/8	1000	---	---	---	---	---	---	---	---	---	---	---	---
1 1/4 to 3	8 1/8	9/32	13/32	5/32	2000	---	---	---	---	---	---	---	---	---	---	---	---
2 to 4 3/4	11 3/8	5/16	15/32	3/16	----	1 1/2	1 1/2	---	3/4	1 1/2	1 1/2	3	3	---	2 1/4	3 9/16	2 3/4
	----	--	---	--	3000	2 1/2	2 1/2	---	---	2	---	4 1/8	4 1/8	---	---	4 1/16	---
4 1/2 to 6 1/4	12 1/8	5/16	15/32	1/4	4000	---	---	---	---	---	---	---	---	---	---	---	---
	----	--	---	--	----	---	---	---	---	2 1/2	---	---	---	---	---	4 5/8	---
4 to 6 1/4	15 1/2	7/16	9/16	1/4	5000	---	---	4	---	3	---	---	---	6 3/16	---	5 1/4	---
	----	--	---	--	----	---	---	---	---	4	---	---	---	---	---	6 3/8	---
6 1/8 to 8 3/4	13 3/4	5/16	15/32	5/16	5000	---	---	---	---	---	---	---	---	---	---	---	---

1/ Designations for hose types are as follows:
A - Cotton rubber-lined hose.
B - Unlined linen hose.
C - Oil-suction and oil-discharge hose.
D - Rubber steam hose.
E - Water-suction hose.
F - Wash-deck and engineers' rubber hose.

Source: https://assist.dla.mil - Downloaded: 2026-02-26T12:41Z
Check the source to verify that this is the current version before use.

3.5.2.1 *Bright chromium plate*[1]. The coating shall be electrodeposited metals consisting of nickel, followed by chromium, the minimum thickness being 0.0002 inch for nickel and 0.00001 inch for chromium.

3.5.2.2 *Bright nickel plate*[1]. The coating shall be electrodeposited nickel, and shall be of not less than 0.0002 inch.

3.5.2.3 *Cadmium plate*. The coating shall be electrodeposited cadmium not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.5.2.4 *Zinc plate*. The coating shall be electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.5.2.5 *Hot phosphating*. The coating shall consist of a chemically produced phosplate followed with a coating of polar type, thin film, rust-preventive compound.

3.5.2.6 *Oxide coating*. The coating shall consist of a chemically produced oxide followed with a coating of polar type, thin film, rust preventive compound.

3.5.2.7 *Enamel*. The enamel shall be of good quality suitable for applying to metal surface. The coating shall not be tacky nor flake off easily. When specified (see 6.1), the wrenches shall be furnished with either olive drab or gray enamel as specified in 3.5.2.7.1 or 3.5.2.7.2.

3.5.2.7.1 *Olive drab*. When olive-drab finish is specified, the surface of the wrench shall be prepared for painting in accordance with any method of TT-C-490. The surface shall then be coated with enamel conforming to TT-E-485, and shall match color No. 24087 of Fed. Std. No. 595. Dry film thickness shall be 1 mil minimum.

3.5.2.7.2 *Gray*. When gray finish is specified, the surface shall be coated with light gray equipment enamel in accordance with Formula 111, type III, class 2 of MIL-E-15090.

3.5.2.8 *Lacquer*. Lacquer coating shall be of good quality suitable for applying to bare metal surfaces to deter rusting in transit or short term storage.

3.6 **Type I, adjustable hook.** The adjustable hook(s) of type I spanner wrenches shall be of steel and heat treated to possess a Rockwell C hardness of not less than 39 nor more than 49. (See 4.4.1.)

3.6.1 *Class* 1, *fixed pivot point*. Clearance between the jaw and the handle clevis shall be sufficient to allow for free pivoting of the jaw without excessive looseness. The steel hinge pin shall be peened or otherwise well secured. Wrenches shall be drop-forged steel with a plain or solid web in the handle. Class 1 wrenches shall conform to table I for the capacity specified (see 6.1), and shall be similar to figure 1.



FIGURE 1.—Type I, class 1, adjustable hook, fixed pivot point, spanner wrenches.

3.6.2 *Class* 2, *variable pivot point*. The class 2 wrench shall have a capacity of from 2 to 12 inches and shall be suitable for heavy turret nuts. The wrench shall consist of a slotted metal tubular handle, bolt and nut, and two interchangeable hooks, one 3/4-inch thick and the other 3/8-inch thick with each hook provided with eleven notes. The handle shall have an over-all length of not less than 22 inches and a weight of not less than 10 pounds. The wrench shall be similar to figure 2.

[1] When chrome or nickel coatings are specified, but are restricted by Government order, another coating, as listed, shall be substituted.

Source: http://assist.dla.mil -- Downloaded: 2021-06-12 22:41Z
Check the source to verify that this is the current version before use.

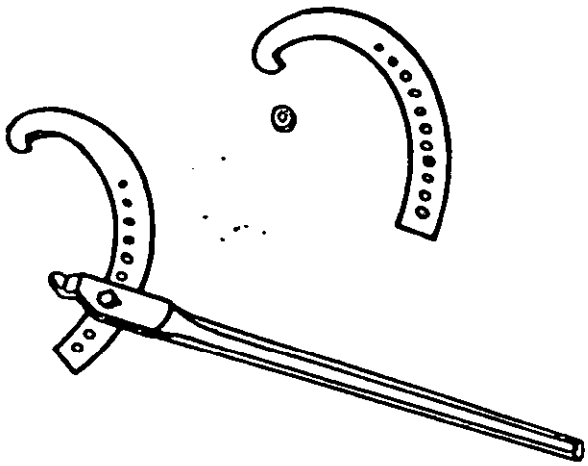


FIGURE 2.—Type I, class 2, adjustable hook, variable pivot point, spanner wrenches.

3.6.3 *Class 3, heavy duty.* The class 3 wrench shall be suitable for use on turret-adjusting nuts, packing-gland nuts, and other turret-type nuts. The wrench shall have a self-adjusting jaw which will provide adjustments from 1-1/2 to 4 inches. The wrench shall have an overall length of not less than 19 inches and a weight of approximately 3 pounds. The wrench shall be drop-forged steel with a plain or solid web in the handle, and shall be similar to figure 3.

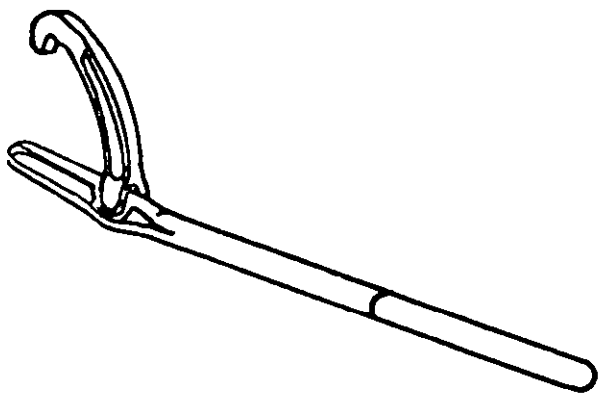


FIGURE 3.—Type I, class 3, adjustable hook, heavy duty, spanner wrench

3.7 **Type II, pin.** Type II wrenches shall have the pins forged integral with the spanner head and shall be machined or be of a separate piece, copper brazed into a reamed hole, or secured in a manner to afford equal retention capacity.

TABLE II. - Type II, grade A, pin spanner wrenches.

Capacity for circle, diameter	Diameter of pin		Pin length +5/64 -1/64	Over-all length (approximate)	Test load (minimum)
	maximum	minimum			
Inches	Inch	Inch	Inch	Inches	Inch-pounds
1	0.187	0.177	1/8	4	200
1 1/4	.203	.192	1/8	4 1/2	500
1 1/2	.218	.207	5/32	5	1000
1 3/4	.234	.222	5/32	5 1/2	1300
2	.250	.237	3/16	6	1600
2 1/4	.265	.253	3/16	6 1/2	1800
2 1/2	.281	.268	7/32	7	2300
2 3/4	.296	.283	7/32	7 1/2	2600
3	.312	.299	1/4	8	3400
3 1/4	.328	.313	1/4	8 1/2	3700
3 1/2	.343	.329	9/32	9	4600
3 3/4	.359	.341	9/32	9 1/2	5000
4	.375	.357	5/16	10	6000
5	.437	.417	3/8	12	7500
6	.500	.478	7/16	14	9500

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.7.1 *Grade A, hardened and tempered steel.* Grade A wrenches shall be of steel, heat treated to possess a Rockwell C hardness of not less than 39 nor more than 49 (see 4.4.1). Wrenches shall be drop-forged with a plain or solid web in the handle, and shall conform to table II for the capacity specified (see 6.1), and shall be similar to figure 4.

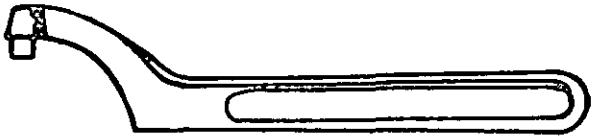


FIGURE 4.—Type II, pin spanner wrenches.

3.7.2 *Grade B, casehardened steel or malleable iron.* Grade B wrenches shall be of casehardened steel or malleable iron, and shall be suitable for use on hole-type couplings for hoses 3/4 and 1 inch in diameter. The pin diameter shall be not less than 1/4 nor more than 5/16 inch. The wrench shall have an overall length of not less than 7 inches and a weight of not less than 8 ounces. The wrench shall be similar to figure 4.

3.8 **Type III, face.** Type III wrenches shall be of steel, shall be drop-forged with a plain or solid web in the handle, and shall be heat treated to possess a Rockwell C hardness of not less than 39 nor more than 49 (see 4.4.1).

3.8.1 *Class 1, adjustable.* Class 1 wrenches shall have the pins forged integral with the handles and shall be machined, or be of a separate piece, copper brazed into a reamed hole, or secured in a manner to afford equal retention capacity. The hinge pin shall be peened or otherwise well secured in a manner to hold the handles in any set position, permit smooth movement and operation. Class 1 wrenches shall conform to table III for the capacity specified (see 6.1), and shall be similar to figure 5.

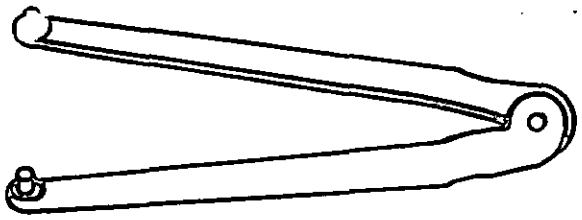


FIGURE 5.—Type III, class 1, adjustable-face spanner wrenches.

3.8.2 *Class 2, nonadjustable.* Class 2 wrenches shall have the pins forged integral with spanner head and shall be machined or be of a separate piece, copper brazed into a reamed hole or secured in a manner to afford equal retention capacity. Class 2 wrenches shall conform to table IV for distance, center to center, as specified (see 6.1), and shall be similar to figure 6.

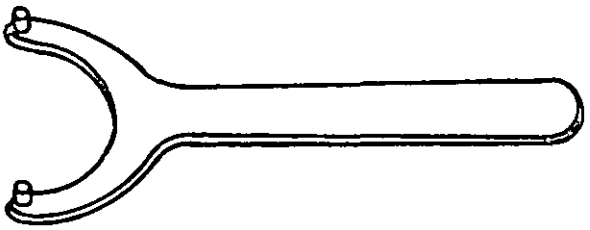


FIGURE 6.—Type III, class 2, nonadjustable-face spanner wrenches.

TABLE III. - Type III, class 1, adjustable face spanner wrenches.

Maximum capacity	Pin diameter		Pin length + 3/32 − 1/64	Over-all length approximate	Test load Minimum
	Maximum	Minimum			
Inches	Inch	Inch	Inch	Inches	Inch-Pounds
2	0.187	0.177	1/8	6 3/8	2,000
3	.250	.237	3/16	8 1/4	3,000
4	.312	.299	1/4	10 3/8	5,000

Source: https://assist.dla.mil - Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE IV. - Type III, class 2, nonadjustable face spanner wrenches.

Pins				Span of jaws in clear, approximate	Length from center of pin to end of handle approximate	Test load minimum
Distance center-to-center ±0.005	Diameter		Length ±1/64			
	Maximum	Minimum				
Inches	Inch	Inch	Inch	Inches	Inches	Inch-Pounds
1	0.187	0.177	3/16	11/16	4 1/2	1200
1 1/4	.218	.207	7/32	7/8	5	1500
1 1/2	.218	.207	7/32	1 1/8	5 7/16	2000
1 3/4	.218	.207	7/32	1 3/8	5 7/8	2500
2	.250	.237	1/4	1 19/32	6 3/8	3000
2 1/4	.250	.237	1/4	1 27/32	6 7/8	4000
2 1/2	.281	.268	9/32	2 1/32	7 1/4	5000
2 3/4	.281	.268	9/32	2 9/32	7 13/16	6000
3	.312	.299	5/16	2 1/2	8 1/4	7000
3 1/4	.312	.299	5/16	2 3/4	9	8000
3 1/2	.312	.299	5/16	3	9 1/2	9000
3 3/4	.375	.357	3/8	3 3/16	10	11300
4	.375	.357	3/8	3 7/16	10 1/2	12400

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.9 **Type IV, combination hydrant and spanner.** Type IV wrenches shall be suitable for use on pin-lug, guard-lug, slot, and rocker-lug-type couplings for hoses of 2, 2-1/2, and 3 inches in diameter. The wrench shall be adjustable and suitable for use on any size hydrant cap up to 1-3/4 inch, pentagon, point to flat, and 1-1/4-inch square nut. The wrench shall have an overall length of not less than 18 inches and a weight of approximately 4-1/2 pounds. The wrench shall be capable of withstanding a test load of 12,500 inch pounds. The wrench shall be either casehardened forged steel or cast malleable iron and shall be similar to figure 7.

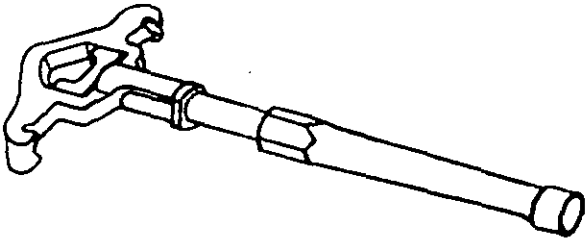


FIGURE 7.—Type IV, combination hydrant and spanner wrench.

3.10 **Type V, universal, hose coupling.** Type V wrenches shall be suitable for use on pin-lug, guard-lug, rocker-lug, and other type couplings for hoses from 1 to 3 inches in diameter. The handle of the wrench shall be shaped so that it can be used as a jimmy or forcible-entry tool and shall be cored to fit standard gas-cock shut-offs. One end of the wrench shall be furnished with a belt-hook eyelet. The wrench shall have an overall length of not less than 11 inches and a weight of not less than 1 pound. The wrench shall be capable of withstanding a test load of 5,000 inch pounds. The wrench shall be either casehardened forged steel or cast malleable iron and shall be similar to figure 8.

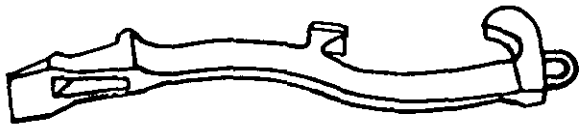


FIGURE 8.—Type V, universal, hose coupling wrench.

3.11 **Type VI, double eye.** Type VI wrenches shall be suitable for use on pin-lug type couplings for hoses 4-1/2 inches in diameter. The wrench shall have an overall length of not less than 16 inches and a weight of not less than 3-1/2 pounds, and be either casehardened forged steel or cast malleable iron. The wrench shall be capable of withstanding a test load of 20,000 inch pounds. Type VI wrenches shall be similar to figure 9.

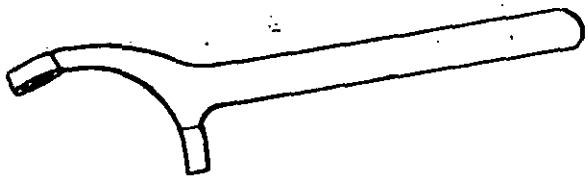


FIGURE 9.—Type VI, double-eye wrench.

3.12 **Type VII, nonadjustable hook for spring grip coupling.** Type VII wrenches shall be of cast malleable iron. The wrench shall be designed for use on 2-1/2-inch spring grip couplings. The wrenches shall conform to table V and shall be similar to figure 10.

TABLE V. - Type VII, non-adjustable hook for spring grip couplings.

Capacity for circle diameter Nominal	Thickness of handle and hook approximate	Length of handle approximate	Overall length approximate	Test load Minimum
Inches	Inch	Inches	Inches	Inch-Pounds
3 11/16	9/32	4 5/8	8 3/16	1,000

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the currentversion before use.

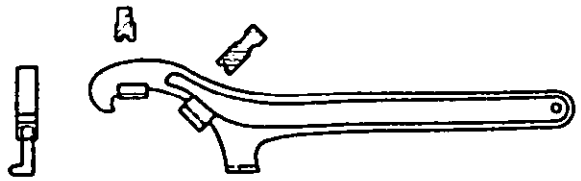


FIGURE 10.—Type VII, nonadjustable hook, for spring grip couplings.

3.13 Workmanship. The workmanship shall be first class throughout. Wrenches shall be free from fins, burrs, external sharp or rough edges, and defects which may impair serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 *Inspection lot.* All wrenches of the same type, size, grade and class presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 *Sampling for examination.* A random sample of wrenches shall be selected from each lot for examination in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300 and level I for lots of 301 and over. The acceptable quality level (AQL) shall be equal to 1.5 percent defective.

4.2.3 *Sampling for hardness and load test.* A random sample of wrenches shall be selected from each lot for test in accordance with MIL-STD-105 at inspection level L8. The acceptable quality level (AQL) shall be equal to 4.0 percent defective. However the sample sizes shall be the number associated with the letters in the table for inspection level L8 and the acceptance number shall be zero until the AQL permits one or greater.

4.3 Inspection.

4.3.1 *Examination.* Each of the sample wrenches selected in accordance with 4.2.2 shall be examined to verify conformance with this specification. Examination shall be conducted as specified in table VI. Any wrench in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective wrenches in any sample exceeds the acceptance number for the sample the lot represented by the sample shall be rejected.

TABLE VI.—*Classification of defects in accordance with Standard MIL-STD-105.*

Categories	Defects
Critical:	None defined.
Major:	
101	Type, class, size and grade and (if applicable) capacity of spanner wrench not as specified.
102	Surfaces not free of cracks, pits or nodules; material not as specified.
103	Hook defective, not clearly defined, or working surfaces not finished as required (type I only).
104	Movable jaw not free pivoting; jaw sticks, binds, or evidences excessive looseness, hinge pin not well secured (type I, classes 1 and 3 only).
105	Hooks not of the required thickness, number of holes not as required, bolt and nut missing, or bolt not of sufficient size (type 1, class 2 only).
106	Depth of hook not within the specified tolerance (type I, class 1 only).
107	Pins defective, not integral with handle or not copper brazed in place (types II and III only).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

Categories	Defects
Critical:	
Major:	
108	Diameter of pins not within the specified maximum and minimum values (types II and III only).
109	Length of pins not within the specified tolerance (type II, grade A and type III, classes 1 and 2 only).
110	Handles stick, bind, or evidence excessive play; hinge pin not well secured (type III, class 1 only).
111	Distance (center to center) between pins not within the specified tolerance (type III, class 2 only).
112	Span of jaws in the clear not as required (type III, class 2 only).
113	Working surfaces not clearly defined or not finished as required (types IV, V and VI only).
114	Wrench not adjustible; adjusting mechanism sticks, binds, or evidences of excessive looseness (type IV only)
115	Handle not shaped so that it can be used as a jimmy, not cored to fit a standard gas-cock shut-off, or one end of wrench not furnished with a belt eyelet (type V only).
Minor:	
201	Surface not free of fins, burrs, rust, rough surfaces, or sharp edges; corners not broken as required.
202	Coating (chrome plate or lacquer) missing, defective, incomplete, or not as specified; coating not smooth, continuous, or not free of pits, blisters, nodules, or uncoated areas.
203	Overall length less than the specified minimum.
204	Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.
205	Weight of wrench not as specified.

4.3.2 *Quality conformance tests.* Each of the sample wrenches selected in accordance with 4.2.3 shall be tested in accordance with the applicable hardness paragraph and the applicable table for test loads to verify conformance with this specification. Any sample wrench which does not meet the requirements for any of these characteristics shall be rejected and if the number of nonconforming wrenches in any one sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 **Test procedures.**

4.4.1 *Hardness tests.* Rockwell hardness tests shall be conducted in accordance with method 243 of Fed. Test Method Std. No. 151. Surfaces shall be suitably ground for testing, particularly to remove the plating and decarburization.

4.4.2 *Test load for type I, class* 1, *type II, grade A, type III, classes* 1 *and* 2 *and types IV, V, VI and VII.* Test loads shall be conducted on the sample wrenches to determine conformance with applicable test load requirements.

4.4.2.1 *Reference lines for determining permanent deformation.* In order to prepare the sample for test, suitable reference lines shall be scribed on the jaws and handles of the wrenches. After the application of the test load the scribed lines shall be examined for permanent deformation.

4.4.2.2 *Test procedure.* The sample wrenches shall be mounted on a suitable mandrel or clamping device and the test load shall be applied either with suitable torque-producing machine lever with dead weight, testing machine or other suitable device. The test load shall be applied as near as practicable to the handle end and in such a manner that the effective lever arm can be measured. The test load shall be the product of the effective lever arm, in inches, and the applied load, in pounds.

4.4.2.3 *Test failure.* Any wrench which shall develop cracks, remain with perceptible deformation, or otherwise develop defects rendering the wrench unserviceable after test, shall be considered as having failed the load test.

4.5 **Inspection for preparation for delivery.** Inspection to determine compliance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

preparation for delivery shall be made in accordance with MIL-H-15424.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Wrenches, preservation and packaging shall be in accordance with level A of MIL-H-15424. Preservative compounds shall not be required for wrenches when surfaces are protected with a vitreous or plastic coating, prime coated, painted, plated or for surfaces that are inherently resistant to corrosion. Unit packaging of these type wrenches for level A shall be in accordance with method III of MIL-P-116.

5.1.2 *Level C.* Cleaning, preservation and packaging shall be in accordance with the supplier's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Wrenches, packaged as specified (see 6.1) shall be packed in accordance with the Level A requirements of MIL-H-15424.

5.2.2 *Level B.* Wrenches, packaged as specified (see 6.1) shall be packed in accordance with the Level B requirements of MIL-H-15424.

5.2.3 *Level C.* Wrenches, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.* In addition to any special marking required by the contract or order herein, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* Marking for shipment shall be in accordance with the standard marking of Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and grade of wrench required (1.2).
(*c*) If a specific color is desired, state requirements (3.5.2.7).
(*d*) Capacity or center to center of pins as applicable (3.6.1, 3.7.1, 3.8.1 and 3.8.2).
(*e*) If a specific coating is required, state type desired (3.5.2).
(*f*) Levels of packaging and packing required (5.1 and 5.2).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Wrenches, not otherwise indexed by name, without attachments.
Carload minimum weight 30,000 pounds.

Motor:

Wrenches, not otherwise indexed, without attachments.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—GL

Navy—Sh

Air Force—MOA

Copies of this specification may be purchased for 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.